

Abhishek Bharadwaj

Bengaluru, Karnataka — 📞 +91-7907424425 — ✉ bharadwajabhishek300@gmail.com
🌐 github.com/abhib03 — 💼 linkedin.com/in/abhishekbharadwaj03

Education

Sri Jayachamarajendra College of Engineering (JSS-STU), Mysuru Nov 2022 – June 2026
Bachelor of Engineering in Electronics and Instrumentation Engineering

Technical Skills

- **Programming Languages:** C, C++
- **Microcontrollers & Platforms:** STM32F411CE (ARM Cortex-M4), ESP32, Raspberry Pi 5, Raspberry Pi Zero 2 W
- **Embedded Systems:** Bare-metal firmware, register-level programming, peripheral drivers, interrupt handling, sensor interfacing, FreeRTOS
- **Peripherals & Protocols:** GPIO, SPI, I²C, UART/USART, ADC, 1-Wire
- **Debugging & Hardware:** ST-LINK/SWD, oscilloscope, logic analyzer, digital multimeter, circuit prototyping, soldering
- **Software & Tools:** STM32CubeIDE, ARM GNU Toolchain, VS Code, Git, GitHub, Linux

Experience

- Software Intern — Aktivolt** July 2026 – Present
- Onboarding into ESP32-S3 firmware development using ESP-IDF v5.4, focusing on peripheral interfacing and embedded C fundamentals.
 - Studying BMS architecture and embedded protocols (CAN, SPI, I2C, UART) relevant to e-mobility applications.
 - Using debugging and testing workflows to investigate firmware and hardware integration issues.

Projects

- Bare-Metal STM32F411xx Device Drivers** Dec 2025 – Present
- Developed reusable register-level peripheral drivers in Embedded C for the STM32F411CE ARM Cortex-M4 without relying on STM32 HAL/LL libraries.
 - Implemented modular **RCC, GPIO, I²C and SPI** drivers using peripheral handles, configuration structures, and reusable driver APIs.
 - Implemented GPIO input/output configuration, pull-up/pull-down control, alternate-function mapping, output control, input reading, toggle operations, and GPIO interrupt configuration support.

- Embedded Acoustic Camera** March 2026 – May 2026
- Developed a real-time embedded acoustic camera using Raspberry Pi 5 and a circular microphone array for 2D sound-source localization.
 - Implemented GCC-PHAT time-difference-of-arrival estimation and a least-squares direction-of-arrival solver for spatial sound localization.
 - Implemented a 64×64 Delay-and-Sum beamformer and integrated the processing pipeline with a live camera feed to generate real-time spatial heatmaps.

- Neonatal Monitoring System** Jun 2025 – Jul 2025
- Developed a real-time monitoring system on Raspberry Pi Zero 2 W using MAX30102, MPU6050, and DS18B20 sensors.
 - Interfaced sensors using I²C and 1-Wire protocols and processed heart rate, SpO₂, temperature, and motion data.

Patent Publication

- Real-Time Embedded System for Visualizing Sound Source Direction and Relative Intensity Using a Circular Microphone Array**
- Indian Patent Application No. **202641070671**; filed 05 Jun 2026 and published 12 Jun 2026 (Journal No. 24/2026).

Achievement

- **Reliance Foundation Scholar** — Reliance Foundation Scholarship, 2022–2026